Summary of Revisions

Interpretable Multivariate Conformal Prediction with Fast Transductive Standardization

JMLR-25-3153-1

This summary compares the revision with the original arXiv v1 submission. It distinguishes the author's edits, added experiments, and the final publication-style reorganization; it does not certify new theoretical results.

- **Abstract and Section 1: motivation and related work.** The central motivation and contribution statement are retained; no substantive rewrite is claimed. **Author check:** Correct the Point CHR/CQHR distinction and add shape-template related work. These changes are necessary for accurate positioning of the new comparisons.
- **Section 2: setup and motivating constructions.** The supplied revision renames and reorganizes the preliminaries, clarifies joint exchangeability and simultaneous versus marginal coverage, strengthens vector/scalar distinctions, discusses ties, and introduces a common parameterized population-oracle formulation. This makes the statistical target and the role of standardization more explicit. The latest edit removes the unqualified shift-invariance claim and replaces "uniformly tight" with a coverage-balance description. **Author check:** Reconcile capped-score ties with the implementation and qualify the remaining balance claim.
- **Section 3: methodology.** The supplied edit uses a common threshold-indexed prediction-set map, separates observed and oracle scaling statistics, rewrites the global/local construction in that notation, and corrects the exceptional empty-cell sign. The latest edit replaces the local conditional-coverage inference with a pointwise oracle-acceptance implication, improving mathematical precision. **Author check:** Review the search-lemma endpoint inequality and complete the common-event argument in the piecewise coverage proof.
- **Section 4: numerical experiments.** New main-text material includes dependent heterogeneous Gaussian noise, coordinate-length bar plots, partial heteroskedasticity, native CQHR, and real-data wall-clock comparisons. New 200-split real-data reruns add all coordinate lengths and direct backward-search/fallback counts. Independent-noise benchmarks now lead into the dependence studies; the original energy example remains in the body. Sampling protocols and CQR settings are explicit. These changes answer requests for coordinate detail, broader baselines, and computational evidence; the Point CHR infinite-volume case is now explained by its allocation.
- **Section 5: discussion.** The existing discussion is retained without alteration. **Author check:** Connect it to the new contamination and CQR evidence while retaining the distinction between empirical robustness and proved validity.
- **Appendix A: algorithms and complexity.** The supplied algorithmic material is retained. No faster worst-case bound is claimed. The new Section 4 diagnostics measure actual search branches and their observed costs, rather than inferring branch frequency from overall runtime.
- **Appendix B: proofs.** The supplied proof section is preserved, including its updated notation. No new moment-relaxation or tie-handling theorem is claimed. **Author check:** Check mathematical consistency with the revised main text; this integration did not perform a complete proof-equivalence audit.
- **Appendix C: unified experimental evidence.** Distribution benchmarks, dependence and heterogeneity sweeps, calibration size, heavy tails, contamination, CQR transformations, shape templates, oracle approximation, computational scaling, and real-data results are now organized by scientific topic. The earlier and added studies jointly show where efficiency gains persist and where they are limited.
- **Experimental presentation and preservation.** The separate legacy appendix has been folded into Appendix C. `experiments_body.tex` and `experiments_appendix.tex` now contain the complete account, without a revision-log narrative. All experimental settings, numerical results, figures, and tables are unchanged by this editorial pass. Fixed-pool and independent-draw protocols remain explicitly distinguished so the reorganization does not obscure their scope.
- **Figures, bibliography, and remaining completion.** All original experimental assets and labels are retained; 19 new PDF figures and the Tumu et al. reference are added. Figures 1 and 2 now show illustrative marginal coverage rates and residual-boundary labels, respectively, to clarify their interpretation. **Author check:** Finish notation consistency and manuscript-wide color marking. The separate reviewer audit lists the remaining items.